1. Outline

This report presents a structured analysis of global disaster data using Tableau. It follows a systematic workflow beginning from data understanding to final insights and recommendations.

- Introduction
- Story of Data
- Data Splitting and Preprocessing
- Pre-Analysis
- In-Analysis
- Post-Analysis and Insights
- Data Visualizations & Charts
- Recommendations and Observations
- Conclusion
- References & Appendices

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2. Introduction

Objective of the Project

The objective of this analysis is to examine global disaster trends between 2018 and 2024 in order to understand patterns in disaster severity, casualties, response efficiency, economic loss, and recovery time. The study also aims to evaluate how preparedness and response systems influence disaster outcomes.

Problem Being Addressed

Disasters continue to cause significant loss of life, economic disruption, and long recovery periods globally. This analysis seeks to answer key questions such as which disaster types are most deadly, how response time affects outcomes, and whether humanitarian aid is effectively distributed.

Key Datasets and Methodology

- Dataset: Global disaster records (2018–2024)
- Tool: Tableau
- Methods: descriptive statistics, trend analysis, correlation exploration, and data visualization (charts and maps)

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3. Story of Data

Data Source

The dataset is derived from global disaster monitoring records covering multiple countries from 2018 to 2024.

Data Collection Process

Data was compiled from disaster reporting systems and humanitarian response databases, capturing both natural and climate-related disasters.

Data Structure

- Rows: Individual disaster events
- Columns: Country, disaster type, casualties, severity index, economic loss, response time, aid amount, recovery duration, and response efficiency score

Key Features and Significance

- Country: Identifies geographic risk distribution
- Disaster Type: Categorizes disaster events
- Casualties & Severity Index: Measures human impact
- Economic Loss (USD): Quantifies financial damage
- Response Time (hours): Measures emergency efficiency
- Aid Amount: Indicates humanitarian support
- Recovery Days: Measures long-term disruption

Data Limitations

- Possible underreporting in low-income regions
- Variations in reporting standards across countries
- Missing or estimated values in some disaster events

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4. Data Splitting and Preprocessing

Data Cleaning

- Removal of duplicate disaster records
- Standardization of country and disaster type labels
- Correction of inconsistent time formats

Handling Missing Values

- Missing numerical values were treated using average imputation
- Critical missing records were excluded when necessary

Data Transformations

- Creation of derived indicators such as response efficiency score
- Grouping disaster types for comparative analysis

Data Splitting

Independent Variables:

- Country
- Disaster Type

Dependent Variables:

- Severity Index
- Casualties
- Economic Loss (USD)
- Response Time (hours)
- Aid Amount
- Response Efficiency Score
- Recovery Duration (days)

Industry Context

Humanitarian and disaster management sector

Stakeholders

- Governments
- Health organizations
- Humanitarian agencies
- Affected communities

Value to Industry

This analysis supports better disaster preparedness, resource allocation, emergency response optimization, and policy development for risk reduction.

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5. Pre-Analysis

Key Trends Identified

- Certain countries consistently experience longer response times
- Some disaster types appear more lethal and disruptive than others
- Casualty levels show noticeable spikes in specific years

Potential Correlations

- Faster response times may reduce casualties
- Higher severity index likely leads to increased economic loss
- Certain disasters appear linked with longer recovery periods

Initial Insights

- Storm-related disasters appear to dominate casualty figures
- Recovery time varies significantly by disaster type
- Aid distribution appears concentrated in highly affected countries

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6. In-Analysis

Analysis 1: Response Time Variation

There is significant variation in emergency response times across countries. Some nations consistently respond faster, while others experience delays, particularly in disaster-prone regions.

Analysis 2: Recovery Periods by Disaster Type

Tornadoes and volcanic eruptions show the longest recovery periods (around 50 days), followed by wildfires, hurricanes, and landslides. These disasters cause extensive infrastructure and livelihood disruption.

Analysis 3: Disaster Types and Casualties

Storm surges, tornadoes, and wildfires are the leading causes of casualties, each contributing over 500,000 casualties on average.

Analysis 4: Trend in Casualties Over Time

Casualties peaked in 2020, indicating a major surge in global disaster impact. Although there was a slight decline afterward, levels remain consistently high.

Analysis 5: Humanitarian Aid Distribution

Brazil received the highest disaster aid funding, followed by Bangladesh and China, reflecting both disaster scale and international support needs.

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7. Post-Analysis and Insights

Key Findings

- 2020 recorded the highest global disaster casualties
- Storm surges, tornadoes, and wildfires are the deadliest disasters
- Tornadoes and volcanic eruptions cause the longest recovery delays
- Response efficiency varies significantly across countries

- High aid allocation does not always correspond to faster recovery

Comparison with Expectations

Initial assumptions that higher aid would directly improve recovery speed were not fully supported. Some highly funded countries still experience long recovery periods.

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8. Data Visualizations & Charts

The analysis included the following Excel visualizations:

- Bar charts showing casualties by disaster type
- Line graphs tracking casualties from 2018–2024
- Heat maps showing response time variation by country
- Column charts comparing aid distribution
- Scatter plots showing relationship between severity and economic loss

Key Interpretation

- Storm-related disasters dominate casualty charts
- 2020 appears as a clear peak year in disaster impact
- Countries with slower response times tend to show higher casualty levels

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9. Recommendations and Observations

Actionable Recommendations

- Increase investment in early warning systems and disaster preparedness
- Strengthen emergency response systems in high-risk countries
- Develop disaster-specific recovery frameworks for high-impact events
- Improve disaster logistics and rapid response coordination
- Align humanitarian aid with long-term resilience building, not just emergency relief

Observations

- Current systems are largely reactive rather than preventive
- High-risk disaster types require specialized recovery strategies
- Some countries remain dependent on external humanitarian aid

Unexpected Outcomes

Despite significant humanitarian aid allocation, recovery time remains prolonged in several countries, indicating inefficiencies in resource utilization.

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10. Conclusion

Key Learnings

- Disaster impact remains consistently high globally
- Response time is a critical factor influencing casualties
- Certain disaster types cause disproportionate damage and recovery delays

Limitations

- Data inconsistencies across countries
- Possible underreporting in developing regions
- Limited granularity in some disaster metrics

Future Research

- Incorporate climate change indicators into disaster prediction models
- Expand dataset to include more recent years beyond 2024
- Develop predictive models for response optimization

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11. References & Appendices

References

- Global disaster monitoring datasets (2018–2024)
- Humanitarian response databases